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REJECTION RATE

$$r(\beta,\beta)=||\pi_\beta\otimes\pi_{\beta'}-\pi_{\beta'}\otimes\pi_\beta||_{\rm TV}$$

- We can extend the rejection rate $r_n = r(\beta_{n-1}, \beta_n)$ for any β, β' ,

$$r(\beta, \beta') = 1 - \mathbb{E} \left[1 \wedge \frac{\pi_\beta(X') \pi_{\beta'}(X)}{\pi_\beta(X) \pi_{\beta'}(X')} \right], \quad (X, X') \sim \pi_\beta \otimes \pi_{\beta'}$$

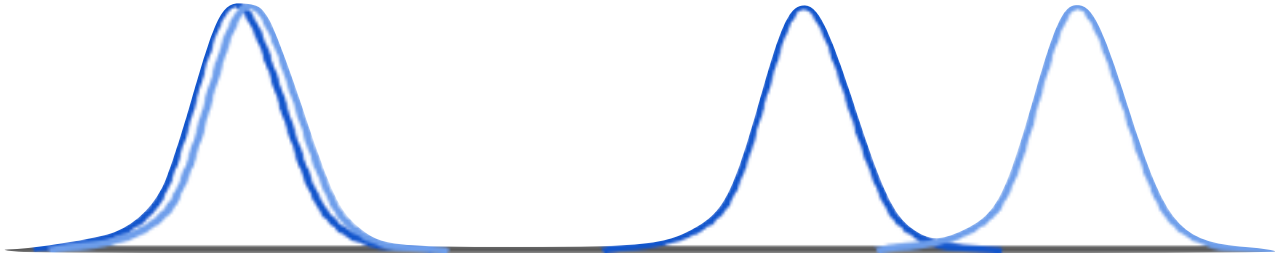
▶ Define a divergent path centered by the overlap between distributions

$$\leq 2||\pi_\beta - \pi_{\beta'}||_TV$$



By Pinsker's inequality we have

$$r(\beta, \beta')^2 \leq \frac{1}{2} \text{KL}(\pi_\beta || \pi_{\beta'}) + \frac{1}{2} \text{KL}(\pi_{\beta'} || \pi_\beta)$$



$$r(\beta, \beta') \approx 0$$

$$r(\beta, \beta') \approx 1$$

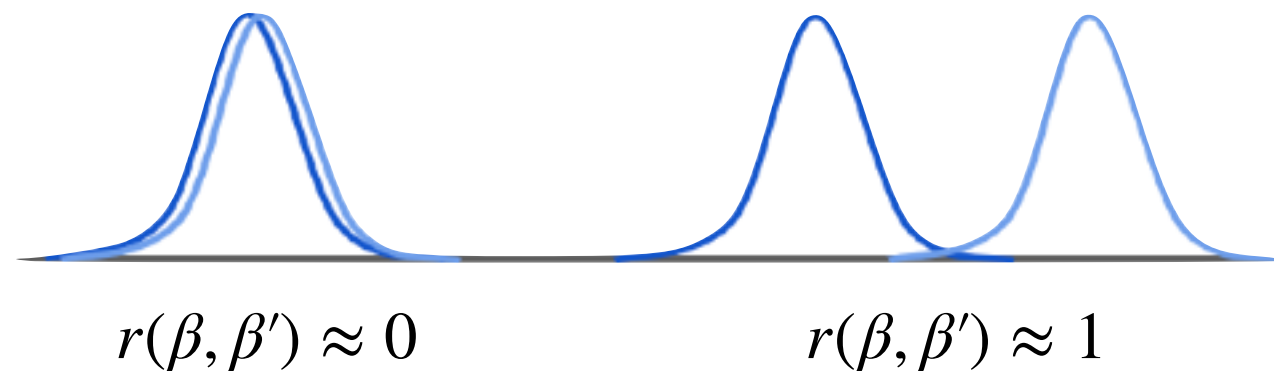
REJECTION RATE

- We can extend the rejection rate $r_n = r(\beta_{n-1}, \beta_n)$ for any β, β' ,

$$r(\beta, \beta') = 1 - \mathbb{E} \left[1 \wedge \frac{\pi_\beta(X')\pi_{\beta'}(X)}{\pi_\beta(X)\pi_{\beta'}(X')} \right], \quad (X, X') \sim \pi_\beta \otimes \pi_{\beta'}$$

- Defines a divergence on the annealing path controlled by the overlap between distributions

$$r(\beta, \beta') = \|\pi_\beta \otimes \pi_{\beta'} - \pi_{\beta'} \otimes \pi_\beta\|_{\text{TV}} \leq 2\|\pi_\beta - \pi_{\beta'}\|_{\text{TV}}$$



- By Pinsker's inequality we have

$$r(\beta, \beta')^2 \leq \frac{1}{2}\text{KL}(\pi_\beta \| \pi_{\beta'}) + \frac{1}{2}\text{KL}(\pi_{\beta'} \| \pi_\beta)$$

LOCAL BARRIER